

End Term Q3

Question 1

```
# def sum_rec(n: int) -> int:
```

```
# if n == 2:
```

```
# return -22
```

```
# else:
```

```
# return -n * (n + 9) + sum_rec(n - 1)
```

```
#
```

```
#
```

```
# def main():
```

```
# lists = []
```

```
# while True:
```

```
# num = int(input())
```

```
# if num == 0:
```

```
# break
```

```
# lists.append(sum_rec(num))
```

```
# for i in lists:
```

```
# print(i, end=" ")
```

```
# return
```

```
#
```

```
#
```

```
# if __name__ == "__main__":
```

```
# main()
```

```
# Question 2
```

```
# def sum_rec(n: int) -> int:
```

```
# if n == 2:
```

```
# return 88
```

```
# else:
```

```
# return n * n * n * (n + 9) + sum_rec(n - 1)
```

```
#
```

```
#
```

```
# def main():
```

```
# lists = []
```

```
# while True:
```

```
# num = int(input())
```

```
# if num == 0:
```

```
# break
```

```
# lists.append(sum_rec(num))
```

```
# for i in lists:
```

```
# print(i, end=" ")
```

```
# return
```

```
#
```

```
#
```

```
# if __name__ == "__main__":
```

```
# main()
```

```
# Question 3
```

```
# def sum_rec(n: int) -> int:
```

```
# if n == 3:
```

```
# return 45
```

```
# else:
```

```
# return n * n * ((n * n) - 4) + sum_rec(n - 1)
```

```
#
```

```
#
```

```
# def main():
```

```
# lists = []
```

```
# while True:
```

```
# num = int(input())
```

```
# if num == 0:
```

```
# break
```

```
# lists.append(sum_rec(num))
```

```
# for i in lists:
```

```
# print(i, end=" ")
```

```
#
```

```
# return
```

```
#
```

```
#
```

```
# if __name__ == "__main__":
```

```
# main()
```

```
# Question 4
```

```
# def sum_rec(n: int) -> int:
```

```
# if n == 4:
```

```
# return 8
```

```
# else:
```

```
# return ((n * n) - (2 * n)) + sum_rec(n - 1)
```

```
#
```

```
#
```

```
# def main():
```

```
# lists = []
```

```
# while True:
```

```
# num = int(input())
```

```
# if num < 0:
```

```
# break
```

```
# lists.append(sum_rec(num))
```

```
# for i in lists:
```

```
# print(i, end=" ")
```

```
#
```

```
# return
```

```
#
```

```
#
```

```
# if __name__ == "__main__":
```

```
# main()
```

```
# Question 5
```

```
# def sum_rec(n: int) -> int:
```

```
# if n == 2:
```

```
# return -1

# else:

# return 1 / 2 * ((n * n * n) - (5 * n)) + sum_rec(n - 1)

#

#

# def main():

# lists = []

# while True:

# num = int(input())

# if num == 0:

# break

# lists.append(sum_rec(num))

# for i in lists:

# print(int(i), end=" ")

#

# return

#

#
```

```
# if __name__ == "__main__":
```

```
# main()
```

```
# Question 6
```

```
# def sum_rec(n: int) -> int:
```

```
# if n == 3:
```

```
# return 54
```

```
# else:
```

```
# return n * n * ((n * n) - 3) + sum_rec(n - 1)
```

```
#
```

```
#
```

```
# def main():
```

```
# lists = []
```

```
# while True:
```

```
# num = input()
```

```
# if str(num) == "STOP":
```



```
# break
```

```
# lists.append(sum_rec(int(num)))
```

```
# for i in lists:
```

```
# print(i, end=" ")
```

```
# return
```

```
#
```

```
#
```

```
# if __name__ == "__main__":
```

```
# main()
```